

SIH Signal Analyzer

Signal Analysis & Intelligence Report

Report Generated: 2026-09-24 00:52:56
Classification: Automated ELINT / SIGINT

EXECUTIVE SUMMARY

RF capture 'synthetic_BPSK_1000kHz_SNR20dB.wav' contains 17,408 samples recorded at a sample rate of 1,000,000 Hz (1,000,000.0 Hz) (effective duration: 17.41 ms (0.01741 s)). The deep-learning classifier evaluated the transmission and identified the modulation as BPSK with 100.0% confidence. Spectral analysis determined an occupied bandwidth of 173.828 kHz (173,828.1 Hz) with a center frequency offset of -4.88 Hz. Demodulation recovered 2,083 raw bits. Frame sync locked to CCSDS_ASM (100% match) with header at bit 110 and payload commencing at bit 142. Channel decoding detected direct (no) interleaving, and raw uncoded framing, successfully assembling 243 decoded payload bytes.

INPUT INFORMATION

Filename	synthetic_BPSK_1000kHz_SNR20dB.wav
Full Path	C:\Sancha\signal-analyzer\synthetic_BPSK_1000kHz_SNR20dB.wav
File Type	WAV Audio / IQ (.WAV)
File Size	69,676 bytes
Sample Rate	1,000,000 Hz (1,000,000.0 Hz)
Total Samples	17,408
Signal Duration	17.41 ms (0.01741 s)

DETECTED PARAMETERS

Modulation	BPSK (100.00%)
Occupied Bandwidth	173.828 kHz (173,828.1 Hz)
Carrier Freq Offset	-4.88 Hz
De-interleaver	none
FEC Scheme	none
Decoded Bytes	243 bytes
Header Offset	110 bits
Payload Offset	142 bits

MODULATION CLASSIFIER DISTRIBUTION

Class	Confidence	Probability Distribution
BPSK	100.00%	
QPSK	0.00%	
GMSK	0.00%	

VISUAL SIGNAL ANALYSIS

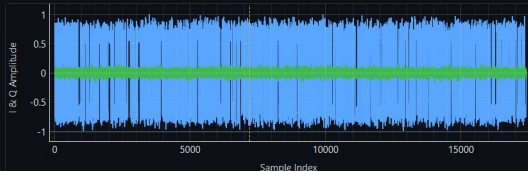


Figure 1: Time-Domain Waveform (In-Phase I & Quadrature Q components)

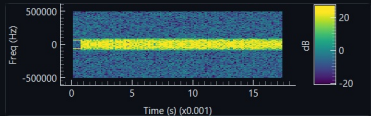


Figure 2: STFT Spectrogram Waterfall



Figure 3: Power Spectral Density (PSD) & Occupied Bandwidth

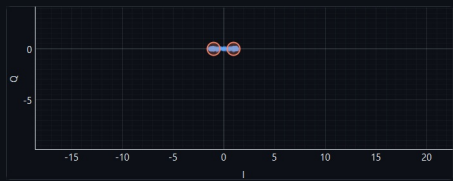


Figure 4: IQ Constellation Diagram with Reference Symbol Overlays

BITSTREAM & FRAME ANALYSIS

Total Bits Analyzed	1,941
Symbol Distribution	954 (49.1%) 987 (50.9%)
Bitstream Sample	10101101 11110000 00101011 11000111 11101000 10111011 00010011 01111101 01010010 01110000 00011110 11100001 01010011 11000111 00111110 00000000 00101011 11011000 01111111 01100111 10001101 00011010 10000101 01001111 10110001 01111110 01000000 00000100 11101000 01011011 11011111 00101111 ... (+1685 bits)
ASCII Text Preview	...+...}Rp.S>+.g.O~@-f/-\$.6d.r1j5.>...Ru../T...~...3..Vnrb.b..

Decoded Bytes Hex Dump:

```
000000  A0 F0 20 C7 E8 88 12 7D 52 7B 1E E1 53 C7 9E 00  |...+...}Rp...S>+|
000010  28 C8 7F 07 89 1A 85 4F 81 7E 4B 04 F4 58 DF 2F  |+...g...O..@-[/|
000020  80 3A 24 FA A4 36 64 85 72 99 9D 6A 35 01 14 3E  ||.$.6d.r1.j5.>+|
000030  C3 DE 18 52 75 F5 EF 2F 54 18 A6 EC E9 7E F3 87  |...Ru.../T...~...|
000040  95 33 81 96 56 6E 72 62 9D 62 9E 94 3C 81 4F 5F  ||.3..Vnrb.b..|
```